Hadoop & Spark - ISEN 2025/2026

Lab: RDDs

For this Lab you will need:

- The 3.1 Lab

As a reminder

Check 3.1 to login in NGROK, Kaggle, install Java or Spark and import libraries.

Manipulate RDDs

Create a SparkContext

```
sc = spark.sparkContext
```

1 Create a RDD using parallelize

```
rddParallel = sc.parallelize(
    [
        [1,"one"],
        [2,"two"],
        [3,"three"],
        [4,"four"],
        [5,"five"],
     ]
    )
```

1 Create a RDD using parallelize

Check RDD partitions

```
rddParallel.getNumPartitions()
```

Get complete result from RDD

```
output = rddParallel.collect()
print(output)
```

1 Create a RDD using parallelize

Get X records from the RDD (3 here)

```
rddParallel.take(3)
```

Get the first element

```
rddParallel.first()
```

1 Create a RDD using parallelize

Please not that in above command you had an array as output, here you have a single element, which happend to be an array.

first() =/= take(1)

```
rddParallel.take(1)
```

2 Create an RDD reading a file

```
# Note: NOT the clean ONE !
rddFile = sc.textFile('lego_sets_and_themes.csv')
rddFile.take(5)
```

We will delete the header as a RDD is just a collection of elements. Note that every line is of type string.

```
header = rddFile.first()
rddFile = rddFile.filter(lambda row: row != header)
rddFile.take(5)
```

2 Create an RDD reading a file

See the number of partitions

```
rddFile.getNumPartitions()
```

You can control the number of partitions by specifying at read time, here 4:

```
# Note: The clean one!
rddFile = sc.textFile('lego_sets_and_themes_clean.csv', 4)

# remove the first row again
header = rddFile.first()
rddFile = rddFile.filter(lambda row: row != header)

rddFile.getNumPartitions()
```

3 Create an RDD from an already existing RDD

You can create a new RDD from the previous one, in order to split columns following the comma. Note that now every line is of type array (and not string)

```
rddFileWithCols = (
    rddFile.map(lambda line: line.split(";"))
)
rddFileWithCols.take(3)
```

Learn more on Lambda functions: here

3 Create an RDD from an already existing RDD

We can also create RDD with other operations such as filters

```
rddFiltered = (
    rddFileWithCols.filter(
        lambda line: line[5] == "Star Wars" # filter on Star Wars sets
        and float(line[2]) >= 2010 # relased after 2010
        )
)
rddFiltered.take(3)
```

Exercice RDD

Using the PySpark RDD `rddFileWithCols`, create a new RDD that filters the Lego sets that belong to the "Harry Potter" theme and were released between the years 2005 and 2010 (inclusive). Display the first 5 elements of the filtered RDD.